National Science Foundation

Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO)

Development of an Artificial Intelligence (AI) Action Plan

REQUEST FOR INFORMATION RESPONSE
CAPABILITIES STATEMENT

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Attn:

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National Science Foundation
Networking and Information Technology Research and Development (NITRD)
National Coordination Office (NCO)

Mr. D'Souza,

Alvarez & Marsal Public Sector Services, LLC (A&M) is pleased to respond to the Request for Information (RFI) for Development of an Artificial Intelligence (AI) Action Plan. Our response is aligned with Executive Order 14179 (Removing Barriers to American Leadership in Artificial Intelligence), which establishes U.S. policy for sustaining and enhancing America's AI dominance to promote human flourishing, economic competitiveness, and national security.

As the world's leading restructuring firm, A&M offers a combination of expertise gained through our embedded leadership model and our corporate experience delivering more than 1,000 commercial and public sector strategic transformation efforts. In support of our clients, we bring a fully integrated commercial and public sector capability utilizing senior-led teams with strong industry, functional and technology experience. We are confident that digital and AI expertise, and commitment to providing strong senior leadership will allow us to accelerate results and deliver implementable strategies.

If you have any questions or would like to schedule a presentation by our team, please contact me at [redacted] or via email at [redacted]

Sincerely,

Edwad Hanapole
Chief AI Officer – Public Sector Services
Managing Director

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1.0 Introduction

Alvarez & Marsal (A&M) is pleased to provide a response to the National Science Foundation's Request for Information (RFI) on the development of an Artificial Intelligence (AI) Action Plan. As a global leader in performance improvement, digital transformation and AI execution, we recognize the critical role AI will play in sustaining U.S. global leadership, enhancing national security, and driving economic growth.

Our approach will support NSF's mission to develop AI action plan and establish robust standards. This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution

2.0 Addressing National AI Priorities: Key Themes & Recommendations

2.1. AI Infrastructure: Hardware, Data Centers, and Compute Resources

To maintain AI leadership, the United States must establish a resilient, high-performance AI infrastructure that can support national security objectives, economic growth, and innovation. The global race for AI supremacy is fundamentally a competition for computational advantage. Nations that control their AI compute infrastructure will lead the next technological revolution.

Recommendation A1: National AI Compute Infrastructure Initiative

The U.S. must invest in domestic semiconductor manufacturing and AI-focused data centers to ensure computational sovereignty. This initiative should include:

- Direct federal funding for leading-edge semiconductor fabrication facilities specialized in AI accelerators and high-performance computing chips
- Tax incentives for private sector investment in domestic AI hardware manufacturing
- Creation of Strategic AI Economic Zones with streamlined permitting, tax benefits, and infrastructure support to accelerate data center construction in regions with abundant renewable energy sources
- Establishment of a Sovereign AI Hardware Consortium bringing together key industry players, government agencies, and research institutions to ensure U.S. independence from foreign supply chains

The Department of Commerce, in coordination with the Department of Defense and Department of Energy, would oversee this initiative. Implementation would follow a three-phase approach: (1) immediate funding for shovel-ready projects at existing semiconductor facilities; (2) medium-term expansion through new fabrication plants in Strategic AI Economic Zones; and (3) long-term research into next-generation semiconductor technologies specifically optimized for AI workloads.

Recommendation A2: Critical Materials Supply Chain Security

Secure access to essential materials for AI hardware manufacturing by:

- Developing a National Critical Materials Strategy focusing on rare earth elements, specialized metals, and other resources needed for AI chip production
- Establishing a Strategic AI Materials Reserve to guard against supply chain disruptions
- Creating international partnerships with allied nations to diversify supply sources and reduce dependence on potentially adversarial countries
- Providing funding for research into alternative materials and recycling technologie

The initiative would be led by a joint task force including the Departments of Commerce, Defense, Energy, and State. The strategy would employ a whole-of-government approach utilizing the Defense Production Act for critical domestic production, diplomatic channels to secure supply agreements with allied nations, and research grants for developing synthetic alternatives and closed-loop recycling systems.

Recommendation A3: Federal AI Compute Framework

The government should establish a federated AI compute infrastructure:

- Integrates existing high-performance computing resources at national laboratories, universities, and defense facilities into a secure, unified grid accessible to authorized researchers and developers
- Implements a tiered access system that provides computing resources based on national priority, with streamlined access for projects related to national security, healthcare innovation, and energy research
- Establishes regional AI compute hubs across the country, ensuring geographical distribution of resources and creating centers of excellence in AI research

The National Science Foundation would lead this initiative in partnership with the Department of Energy's national laboratories and the Department of Defense. The framework would build upon existing investments like the National Research Cloud pilot program but scale it dramatically to meet the exponential growth in AI computing requirements.

Recommendation A4: AI Compute Access Program

Democratize access to high-performance computing for AI development through:

- Creation of an "AI Compute Credit" system for qualified startups, researchers, and small businesses to access federal compute resources
- Allocation of federal AI compute capacity for commercial applications that advance national economic priorities
- Development of streamlined procurement vehicles to allow government agencies to rapidly acquire AI computing resources from pre-approved vendors

The program would be administered by the National Science Foundation with an interagency oversight committee. Implementation would include a merit-based application process with expedited review cycles, integration with existing research grant mechanisms, and public-private partnerships. The program would measure success through metrics including patents filed, startups launched, and peer-reviewed publications resulting from compute access.

Recommendation A5: Next-Generation Computing Research

Invest in transformative computing technologies:

- Accelerate quantum computing research focused on quantum machine learning algorithms and hardware
- Fund development of neuromorphic computing architectures that mimic biological neural networks to achieve orders-of-magnitude improvements in energy efficiency
- Support research into photonic computing for AI applications, leveraging light-based processing to reduce energy consumption and increase processing speeds
- Establish a Moonshot Challenge program with substantial prizes for breakthroughs in AI computing efficiency

A new DARPA-like Advanced Computing Research Agency would coordinate the initiative, emphasizing public-private partnerships, university research centers, and milestone-based funding. The program would be structured around "grand challenges" with financial incentives for achieving specific performance benchmarks, and increased investment as technologies demonstrate viability.

2.2. AI Model Development and Open-Source Innovation

AI innovation must be balanced between open-source development for accessibility and proprietary advancements for national security and economic competitiveness. The right policy approach can foster both a robust innovation ecosystem and strategic advantages in critical applications.

Recommendation B1: Strategic AI Research Program

Establish a coordinated national AI research initiative with funding of $25 billion over five years:

- Create domain-specific AI research programs in critical sectors including defense, healthcare, energy, agriculture, and financial technology
- Fund a network of AI Centers of Excellence at top research universities, with each center focusing on specific aspects of AI (foundation models, reinforcement learning, explainable AI, etc.)
- Implement a fast-track grant approval process for high-priority AI research, reducing the time from application to funding to under 60 days

The White House Office of Science and Technology Policy would coordinate this initiative across agencies including NSF, NIH, DOE, DOD, and USDA. Each agency would establish AI research offices with streamlined funding mechanisms and domain expertise. Implementation would emphasize coordinated but distributed research investment, avoiding duplication while ensuring comprehensive coverage of strategic AI research areas.

Recommendation B2: Long-Term AI Research Funding

Support fundamental AI research with stable, multi-year funding mechanisms:

- Establish a National AI Fellowship program supporting 1,000 doctoral students and 500 postdoctoral researchers annually in AI-related fields
- Fund interdisciplinary research initiatives combining AI with other fields such as materials science, biotechnology, and quantum physics
- Develop a "Manhattan Project for AI" focused on achieving artificial general intelligence (AGI) with strong security and alignment guarantees

The National Science Foundation would administer this program in coordination with other research agencies. Implementation would include creating new funding mechanisms specifically designed for long-term, high-risk AI research; establishing a prestigious fellowship program with competitive stipends and research budgets; developing interdisciplinary research centers that bring together experts from multiple fields.

Recommendation B3: National AI Testbed Network

Create a nationwide network of regulatory ‘sandboxes’ with an initial investment of $5 billion:

- Establish sector-specific testbeds for healthcare, financial services, transportation, and energy applications of AI
- Implement streamlined regulatory frameworks that allow for rapid experimentation while ensuring basic safety standards

- Create a "Safe Harbor" for companies testing AI systems within approved testbeds, providing liability protections for good-faith innovation
- Develop standardized testing protocols and evaluation metrics to assess AI system performance, safety, and reliability

Coordinated by the Department of Commerce with sector-specific agencies (HHS, DOT, DOE, etc.), implementation would establish testbed environments, approval processes, evaluation protocols, and balanced governance committees.

Recommendation B4: Real-World AI Testing Framework

Establish guidelines and infrastructure for testing AI systems in realistic environments:

- Create dedicated testing environments that simulate real-world conditions for autonomous vehicles, industrial robots, and other physical AI systems
- Develop standardized scenarios and stress tests to evaluate AI system performance under various conditions
- Implement a graduated testing approach that moves from simulation to controlled physical environments to limited real-world deployment
- Establish clear criteria for advancing AI systems through progressive testing stages

NIST would lead the development of testing standards with implementation by relevant regulatory agencies. The framework would be developed through a multi-stakeholder process involving standards organizations, industry representatives, safety experts, and academic researchers.

Recommendation B5: National Open-Source AI Initiative

Invest in developing and maintaining secure, high-quality open-source AI models:

- Fund the development of government-backed foundation models in critical areas such as biomedical research, materials science, and climate modeling
- Establish a security certification program for open-source AI models to ensure they meet minimum standards for safety and security
- Create a national repository of verified, secure AI components and building blocks that developers can incorporate into their applications
- Provide ongoing maintenance funding for critical open-source AI infrastructure to ensure long-term viability

The National Science Foundation would lead this initiative in partnership with national laboratories, academic institutions, and industry consortia. Implementation would include competitive grants for development of open-source models, funding for ongoing maintenance and security updates, establishment of model governance frameworks, and creation of a dedicated evaluation team to assess model performance and security.

Recommendation B6: AI Model Security Framework

Develop comprehensive security standards for AI models:

- Create a mandatory security review process for AI models deployed in critical infrastructure or government applications
- Establish a National AI Security Testing Laboratory to conduct independent evaluations of AI system security

- Develop security standards for training data collection, model architecture, and deployment environments
- Implement a vulnerability disclosure program and bug bounty system for identifying and remediating security flaws in AI systems

NIST, in coordination with other agencies, would lead the development of AI security standards. Implementation would include developing security requirements, establishing a certification process, creating a national laboratory for security research and testing, and implementing a vulnerability disclosure process. The framework would be regularly updated.

2.3. AI Security: Cybersecurity and AI Resilience

AI models must be hardened against cybersecurity risks to prevent vulnerabilities that could be exploited by adversarial actors. As AI systems become more powerful and widespread, ensuring their security becomes a national security imperative.

Recommendation C1: National AI Security Framework

Develop comprehensive security standards for AI systems with binding requirements for critical sectors:

- Establish mandatory security controls for AI systems deployed in 16 critical infrastructure sectors as defined by CISA
- Create a tiered security classification system for AI models based on their capabilities and potential risks
- Develop specialized security requirements for large language models, vision systems, and autonomous decision-making systems
- Require regular security assessments and penetration testing of high-risk AI systems

NIST, with stakeholder input, would lead a phased implementation: (1) baseline standards; (2) sector-specific extensions; (3) voluntary adoption with incentives; (4) mandatory requirements for critical infrastructure and government.

Recommendation C2: AI Supply Chain Security

Implement controls to ensure the integrity of AI development and deployment:

- Establish a Secure AI Development Lifecycle framework with security controls at each stage from data collection to deployment
- Create a trusted vendor program for AI components and services used in government and critical infrastructure
- Implement provenance tracking for datasets, models, and software components used in AI systems
- Develop security standards for AI development environments and model hosting platforms

The Cybersecurity and Infrastructure Security Agency (CISA) would lead this initiative in partnership with NIST and sector-specific regulatory agencies. Implementation would include developing guidelines for secure AI development practices, establishing a certification program for AI components used in government applications, creating technical standards for provenance tracking and verification, and implementing a continuous monitoring program for supply chain risks.

Recommendation C3: National AI Cyber Shield

Invest in AI-powered cybersecurity capabilities:

- Deploy AI-based threat detection and response systems across federal networks and critical infrastructure
- Establish a National AI Cybersecurity Operations Center to coordinate automated responses to cyber threats
- Develop AI systems capable of autonomously hunting for vulnerabilities in critical software and infrastructure
- Create a Cyber Defense AI Competition program to incentivize development of advanced defensive AI capabilities

The Department of Homeland Security and Department of Defense would jointly lead this initiative. Implementation would include developing and deploying AI-based security tools across federal networks, establishing a dedicated security operations center with advanced AI capabilities, creating a federated threat intelligence system that shares data across agencies and critical infrastructure sectors, and launching a series of competitions to accelerate innovation in defensive AI.

Recommendation C4: Adversarial AI Research Program

Fund research into adversarial attacks and defenses for AI systems:

- Establish a dedicated research program focused on identifying and mitigating vulnerabilities in AI systems
- Create a National Red Team for AI that conducts ongoing assessments of critical AI systems
- Develop standardized testing methodologies for evaluating AI system robustness against adversarial attacks
- Fund research into novel attack vectors specific to different types of AI systems and applications

DARPA and the National Science Foundation would co-lead this research program. Implementation would include establishing a network of academic research centers focused on adversarial AI, creating a standing red team composed of leading security researchers from government, academia, and industry, developing standardized benchmarks and evaluation methodologies for AI security, and hosting regular competitions to identify novel attack vectors.

Recommendation C5: AI Model Integrity Framework

Establish comprehensive controls to ensure AI model security:

- Develop standards for secure model training and validation processes
- Implement cryptographic verification of model origins and integrity
- Create a National AI Model Registry for critical applications with verified security properties
- Establish requirements for continuous monitoring of deployed models for signs of tampering or degradation

NIST would lead the development of technical standards with implementation coordinated through relevant regulatory agencies. Implementation would include developing guidelines for secure model development and validation, creating technical standards for model verification and integrity checking, establishing a registry system for models used in critical applications, and implementing requirements for ongoing monitoring and verification of deployed models. The framework would be designed to be technically feasible for current systems while establishing aspirational goals for future development.

Recommendation C6: Training Data Security

Protect the integrity and quality of data used in AI development:

- Establish guidelines for secure data collection, storage, and preparation for AI training
- Create a framework for detecting and preventing data poisoning attacks
- Develop standards for data diversity and quality to ensure robustness
- Implement controls for secure data sharing between organizations for AI development

NIST would develop technical standards in collaboration with industry and academic experts. Implementation would include creating guidelines for secure data collection and preparation practices, developing tools and methodologies for detecting potential poisoning or manipulation, establishing minimum standards for data diversity and representation in different application domains, and creating frameworks for secure data sharing and federation. The initiative would emphasize both technical controls and governance processes to ensure data integrity throughout the AI development lifecycle

2.4. AI in Government: Efficiency, Workforce, and Citizen Services

AI must be integrated across government agencies to improve efficiency, service delivery, and national competitiveness. The federal government can both improve its own operations and serve as a model for responsible AI adoption.

Recommendation D1: Federal AI Transformation Initiative

Launch a comprehensive program to modernize government operations with AI:

- Establish a $10 billion fund specifically for AI implementation across federal agencies
- Create agency-specific AI implementation roadmaps with measurable efficiency targets
- Implement a shared services model for common AI applications (document processing, workflow automation, etc.)
- Develop standardized procurement vehicles and contracting templates for AI services

The Office of Management and Budget would lead this initiative in partnership with the General Services Administration and agency CIOs. Implementation would include establishing a central fund with specific allocation for high-impact AI projects, developing standardized implementation playbooks for common use cases, creating shared service offerings that agencies can adopt without duplicating development efforts, and establishing a Federal AI Center of Excellence to provide technical assistance and share best practices. Success would be measured through quantifiable metrics including cost savings, processing time reductions, error rate improvements, and customer satisfaction scores.

Recommendation D2: Mission-Critical AI Applications

Identify and fund high-impact AI implementations:

- Deploy AI-powered fraud detection systems across benefit-administering agencies (SSA, CMS, VA, etc.) with potential savings of $50 billion annually
- Implement AI-based document processing systems for immigration applications, tax processing, and regulatory filings
- Develop predictive maintenance systems for federal infrastructure, military equipment, and transportation assets
- Create AI-powered emergency response coordination systems for FEMA and other disaster response agencies

The transformation would be led by a new Federal AI Implementation Task Force comprising agency CIOs, domain experts, and industry advisors. Implementation would include identifying high-value applications across agencies, developing standardized technology solutions that can be customized for agency-specific needs, establishing cross-agency working groups to share best practices and lessons learned, and creating a phased implementation roadmap with early wins to demonstrate value. The initiative would emphasize measuring and publicizing outcomes to build support for broader AI adoption.

Recommendation D3: Citizen-Facing AI Services

Transform government-citizen interactions through AI:

- Deploy AI assistants across federal agency websites to provide 24/7 support for citizen inquiries
- Create a unified Federal Services AI Platform that provides a consistent interface across all government services
- Implement voice-enabled AI services for citizens with limited digital literacy or disabilities
- Develop personalized service delivery based on citizen needs and preferences

Citizens interact with government services at critical moments in their lives. During health crises, financial hardship, natural disasters, or major life transitions. Beyond efficiency gains, AI-powered services can make government more accessible to all citizens, including those with disabilities, limited English proficiency, or limited digital literacy.

3.0 Public-Private Collaboration: The Role of Public-Private Partnerships & Value-for-Money Framework in the "Age of AI"

3.1. The Strategic Imperative for AI-Enhanced Public-Private Partnerships

Public-Private Partnerships (PPPs) represent a critical mechanism for advancing America's AI dominance while optimizing government resources and fostering private sector innovation. Traditional PPPs have demonstrated effectiveness in infrastructure development, but AI demands a new generation of collaborative frameworks. These partnerships are specifically designed to address the unique challenges of AI development, including massive computational requirements, data access needs, talent scarcity, and the rapid pace of technological change.

PPPs in the Age of AI need to evolve from traditional arrangements in several ways:

1. Accelerated Development Cycles: Modern PPPs incorporate agile development methodologies, allowing for rapid prototyping, testing, and deployment cycles measured in months; not years.
2. Dynamic Risk Allocation: Given the evolving nature of AI technology, AI-Enhanced PPPs incorporate flexible risk-sharing mechanisms that adjust as technologies mature and new challenges emerge.
3. Outcomes-Based Success Metrics: Rather than focusing solely on project completion milestones, AI-Enhanced PPPs track impact metrics like processing time reductions, decision quality improvements, and return on investment.

The White House should issue an Executive Order directing federal agencies to prioritize AI-Enhanced PPPs in their technology modernization efforts, with specific guidance on flexible procurement mechanisms, accelerated approval processes, and risk-sharing models appropriate for AI initiatives

3.2. Priority Areas for AI-Enhanced PPPs

We recommend prioritizing AI-Enhanced PPPs in the following high-impact domains:

Recommendation E1: National AI Research Infrastructure

The computational requirements for developing cutting-edge AI models have increased exponentially, with training costs for frontier models now exceeding $100 million. This scale of investment is prohibitive for most research institutions and startups, creating a competitive advantage for large technology companies and well-funded foreign competitors. A national research infrastructure would democratize access to computing resources, accelerate innovation across sectors, and ensure U.S. leadership in next-generation AI development.

We recommend establishing a National AI Compute Consortium that:

- Creates a network of government-industry shared supercomputing facilities specifically optimized for AI workloads, with specialized hardware accelerators that deliver 10-100x performance improvements over general-purpose computing systems
- Implements tiered access models allowing researchers from academia, government, and qualifying small businesses to access computational resources based on project merit and national priority alignment, with subsidized rates for projects advancing national security or economic competitiveness
- Develops specialized hardware accelerators and novel computational architectures optimized for next-generation AI models, including neuromorphic computing, optical AI accelerators, and quantum-inspired systems

The Department of Energy's national laboratories, in partnership with the National Science Foundation, should establish a coordinating council with leading AI hardware companies, cloud providers, and research institutions. This council would define technical specifications, coordinate resource allocation, and oversee development of a unified access portal for the distributed computing resources.

Recommendation E2: AI-Driven Critical Infrastructure Protection

Critical infrastructure systems including power grids, transportation networks, and water management systems represent both prime opportunities for AI optimization and significant security vulnerabilities. Recent cyber-attacks have demonstrated the vulnerability of these systems, with potential disruptions costing billions and threatening national security. AI offers unprecedented capabilities to detect anomalies, predict vulnerabilities, and automate responses at a speed and scale impossible for human operators.

We recommend:

- Deploying AI-powered anomaly detection systems across critical infrastructure networks to identify and respond to cyber threats in real-time, capable of processing millions of system events per second to identify subtle patterns indicating potential attacks
- Implementing AI-enhanced predictive maintenance solutions to reduce system failures and extend infrastructure lifespan, leveraging multimodal data from sensors, operational logs, and external factors to anticipate failures before they occur
- Creating autonomous system restoration capabilities that can rapidly respond to physical or cyber disruptions, minimizing downtime and preventing cascading failures across interconnected systems

The Department of Homeland Security (DHS) and the Department of Energy (DOE) should jointly establish a Critical Infrastructure AI Security Initiative, partnering with infrastructure operators and technology companies to develop reference architectures, deployment guides, and evaluation frameworks for AI security solutions across critical sectors.

Recommendation E3: AI Workforce Development

The severe shortage of AI talent represents one of the most significant constraints on U.S. AI leadership. Current estimates indicate a shortage of over 300,000 AI-skilled workers nationwide, with competition for talent driving salaries to unsustainable levels for many organizations. This talent gap particularly affects government agencies, small businesses, and critical infrastructure operators who cannot match the compensation packages offered by large technology companies.

We recommend:

- Creating an AI Skills Certification Framework that establishes recognized credentials for AI professionals at various proficiency levels, from technical implementers to strategic leaders, providing a clear pathway for career advancement and skill development
- Expanding the National Science Foundation's AI Research Institutes program to include intensive AI training programs for mid-career professionals, focusing on practical implementation skills and domain-specific applications rather than purely theoretical knowledge
- Establishing tax incentives for companies that invest substantially in AI workforce development programs, with enhanced incentives for programs targeting underrepresented communities, veterans, and displaced workers from traditional industries

The Department of Labor, in partnership with the National Science Foundation and industry associations, should establish a National AI Skills Coalition tasked with developing standardized curricula, certification programs, and workforce transition pathways. This coalition would coordinate with community colleges, universities, and online learning platforms to rapidly scale training capacity.

Recommendation E4: Sovereign AI Model Development

Foundation AI models increasingly represent critical digital infrastructure, with applications spanning healthcare, financial services, defense, and government operations. Current development of these models is dominated by a small number of large technology companies and foreign entities, creating potential dependencies and vulnerabilities in applications critical to national security and economic resilience.

We recommend:

- Establishing a consortium of U.S. companies and research institutions to develop sovereign, U.S.-controlled foundation models for strategic applications, ensuring that critical AI capabilities remain under U.S. jurisdiction and aligned with American values
- Creating secure testing environments for evaluating model capabilities and vulnerabilities prior to deployment, allowing rigorous assessment without exposing sensitive capabilities or weaknesses
- Developing specialized AI models optimized for government applications with enhanced security and explainability features, prioritizing trustworthiness and alignment with legal and ethical requirements over raw performance metrics

DARPA, in collaboration with the National Security Council and leading research universities, should coordinate a multi-year initiative to develop, evaluate, and deploy sovereign AI models across priority government applications. This initiative should leverage existing research programs while establishing new partnership models with industry.

3.3. Applying the Value-for-Money Framework to U.S. AI Leadership Imperatives

Proper evaluation of AI investments requires a systematic framework that captures both quantitative returns and qualitative benefits while accounting for the unique characteristics of AI technologies. Traditional return-on-investment calculations often fail to capture the full value of AI implementations, leading to underinvestment in high-potential applications and misallocation of resources across competing priorities.

A Value-for-Money (VfM) Framework provides a structured method for evaluating AI investments based on efficiency, transparency, and national strategic objectives.

Figure 1: Value-for-Money Framework for evaluating U.S. National AI investments

VfM Framework Component	Application to U.S. AI Leadership Imperatives
Real-Time Cost-Benefit Analysis	AI project funding should prioritize high-ROI use cases like AI-powered fraud detection in Medicare/Medicaid and cybersecurity for national defense.
Performance-Linked Payments	AI vendors in PPPs should be incentivized through milestone-based funding, ensuring government AI initiatives deliver measurable impact.
Risk Allocation Optimization	AI-related risks (e.g., cybersecurity threats, bias in AI models) should be assigned to the entities best equipped to mitigate them—ensuring national security remains a government function while industry drives innovation.
AI-Enhanced Public Impact Measurement	AI investments should be tracked through citizen impact metrics, such as improved service delivery times in healthcare, reduced energy consumption, and faster response times in disaster relief.
Continuous Monitoring & Adaptive Governance	AI PPP projects should be continuously evaluated using real-time AI-driven performance analytics, allowing for rapid course corrections.

This VfM approach de-risks public AI investments, ensuring taxpayer funds are used effectively and that AI investments align across five critical dimensions:

Real-Time Cost-Benefit Analysis

Traditional cost-benefit analyses often fail to capture the exponential nature of AI capabilities and the network effects of successful implementations. Static analyses conducted at project initiation become rapidly outdated as AI systems evolve, data accumulates, and use cases expand. Real-time analysis ensures investment decisions reflect current system performance and emerging opportunities.

A real-time approach incorporates:

- Dynamic ROI Modeling: AI investments often demonstrate increasing returns as data accumulates and models improve. to continuously refine ROI projections based on observed performance data, adjusting projections as systems mature and capabilities evolve.

- Multi-Factor Benefit Quantification: Quantify benefits such as improved decision quality, reduced risk exposure, enhanced innovation capacity, and strengthened competitive positioning.
- Externality Accounting: Incorporate positive and negative externalities, including environmental impacts, privacy implications, and broader economic effects.

OMB should update Circular A-94 (Guidelines for Cost-Benefit Analysis) to incorporate AI-specific valuation methodologies, including guidance on quantifying decision quality improvements, risk reduction benefits, and network effects resulting from AI implementations.

Performance-Linked Payment Structures

Traditional government contracting models incentivize technology delivery rather than outcome achievement, leading to systems that technically meet specifications but fail to deliver real-world value. AI projects are particularly vulnerable to this misalignment, as technical performance metrics may correlate weakly with operational impact. Performance-linked payment structures ensure that vendors and government agencies maintain focus on valuable outcomes rather than technical specifications.

Our framework recommends:

- Outcome-Based Contracts: Structure vendor contracts around specific, measurable performance improvements (e.g., fraud detection rates, processing time reductions) rather than technology delivery milestones. This approach aligns vendor incentives with government objectives and shifts performance risk to the party best positioned to manage it.
- Sliding Scale Incentives: Implement payment structures that increase vendor compensation when outcomes exceed targets and reduce payments when targets are missed. This approach creates continuous improvement incentives rather than binary success/failure criteria.
- Long-Tail Performance Bonuses: Include incentives for sustained performance improvements that accrue as AI systems gather more data and improve over time. This approach encourages vendors to design self-improving systems rather than static solutions requiring constant manual updates.

The Federal Acquisition Regulation (FAR) should be updated to establish AI-specific contract vehicles that facilitate outcome-based payments, including multi-year performance measurement and flexible payment structures that can adapt to evolving system capabilities.

3.4. Implementing Strategic AI Economic Zones

AI innovation thrives in ecosystems that combine specialized infrastructure, regulatory clarity, talent concentration, and collaborative frameworks. The distributed nature of U.S. innovation creates strengths through diversity but can impede the critical mass necessary for breakthrough advances. Strategic AI Economic Zones create focused environments where resources, talent, and policy innovations converge to accelerate development in priority domains.

To accelerate the development and deployment of AI-Enhanced PPPs, we recommend establishing Strategic AI Economic Zones (SAEZs) that provide concentrated environments for AI innovation with specialized regulatory frameworks, infrastructure investments, and collaborative ecosystems.

Each SAEZ would feature:

- Streamlined Regulatory Frameworks: Tailored regulatory environments that maintain appropriate safeguards while expediting AI development and testing.
- Shared Infrastructure: Pre-built computational resources, data management systems, and physical facilities optimized for AI research and deployment.
- Talent Concentration: Educational partnerships, visa programs, and incentive structures to attract and retain top AI talent.
- Implementation Strategy: The Department of Commerce, in partnership with state governments and private sector leaders, should establish a SAEZ development program with federal matching funds for qualified zones, regulatory coordination assistance, and specialized visa allocations for participating organizations.

We recommend establishing five initial SAEZs, each with distinct specialization:

AI Innovation Zones

Five specialized AI innovation hubs across the United States, each with distinct focus areas:

1. National Security AI Zone (Northern Virginia): Defense, intelligence, and critical infrastructure applications in partnership with DoD, intelligence agencies, and defense contractors.
2. Healthcare AI Zone (Boston/Cambridge): Medical diagnostics, drug discovery, and healthcare delivery optimization in collaboration with teaching hospitals, pharmaceutical companies, and healthcare technology firms.
3. Financial Services AI Zone (New York): Financial risk management, fraud detection, and market analysis through partnerships with financial institutions, regulatory agencies, and fintech startups.
4. Manufacturing and Supply Chain AI Zone (Detroit/Pittsburgh): Industrial automation, predictive maintenance, and supply chain resilience in cooperation with manufacturing companies, logistics providers, and industrial engineering programs.
5. Agricultural and Climate AI Zone (California Central Valley): Sustainable agriculture, resource management, and climate adaptation in partnership with agricultural producers, environmental agencies, and climate technology companies.

Congress should authorize the Department of Commerce to establish a Strategic AI Economic Zone program, with appropriations for matching grants, regulatory coordination staff, and specialized visa programs for qualifying AI professionals and students.

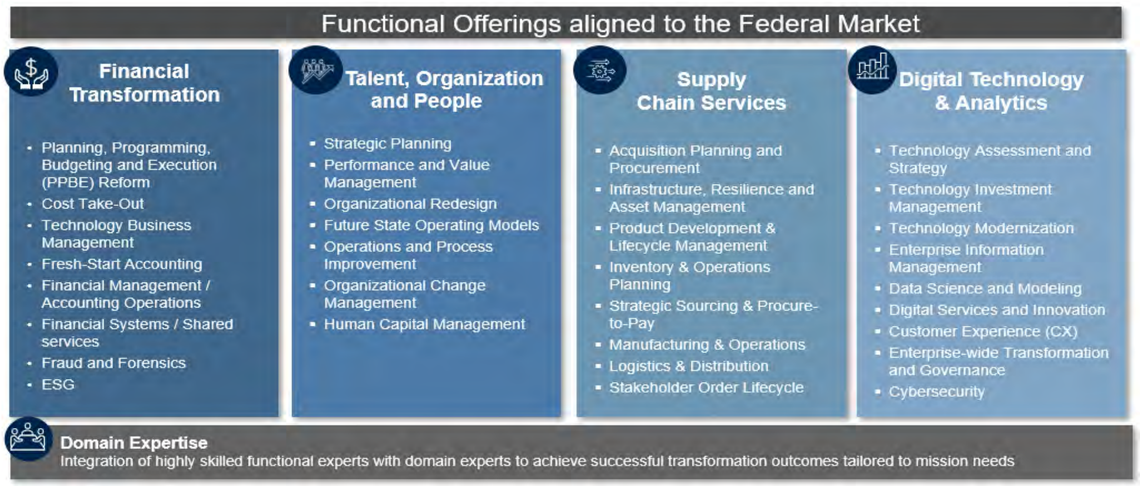
4.0 Capabilities Statement

Alvarez & Marsal Capabilities & Experience

Alvarez & Marsal (A&M) is a global professional services firm with 7500+ employees in over 65 cities and 35 countries.

A&M assists organizations with complex business issues, operating performance, and stakeholder value, specializing in restructuring and turnaround management. A&M provides performance improvement, business transformation, and change management advisory services, with a dedicated Public Sector practice that uses private sector principles to help public sector entities. A&M's Public Sector professionals average 15 to 25 years of experience and include former government and corporate operators and consultants, providing practical, implementable solutions. We combine this expertise with the following tenets to truly drive innovation and transformation for our clients:

Figure 2 -A&M Federal Market Offerings



Alvarez & Marsal Public Sector Services (PSS)

PSS has extensive experience in working across all levels of government from Federal, State and Municipalities. With dedicated Federal, State & Local, Education and Health and Human Services practices, A&M has experience in resolving the most challenging problems for all levels of government. 25% of A&M professionals have held CIO, CTO, CFO and CHRO appointments within government or the private sector. Our teams have been in your shoes, we have experienced the challenges that come from implementing transformative change across governments, in resource constrained environments with missions that cannot fail.

A&M's Strategic Partnership with Virtus

A&M's partnership with Virtus combines A&M's industry expertise and operational excellence with Virtus's technology platform and specialized capabilities. Clients benefit from reduced implementation timeframes, lower total cost of ownership, improved outcomes, scalable solutions, and continuous innovation. The partnership has demonstrated substantial client impact, including an average 30% reduction in implementation timelines, 25-40% improvement in process efficiency, and ROI exceeding initial projections by 15-20%. A&M is committed to delivering exceptional value through this partnership.